Lecture 8

Root strings and rank-one subalgebras

Objectives. *Each root α determines a copy of $\mathfrak{sl}_2$ inside $\mathfrak{g}$. We apply Lecture 6 to $\mathfrak{g}$ as a module for this copy.*

By the end of this lecture you will be able to:

- Construct the root triple of a root and compute its coroot in a matrix algebra.
- Prove that root spaces are lines, that $\pm\alpha$ are the only roots proportional to α, and that $\beta(h_\alpha) \in \mathbb{Z}$ for all roots α, β.
- Determine a root string and compute a Cartan entry from it.
- Explain why the real span of the roots is Euclidean, while the Killing form of $\mathfrak{g}$ is not.

8.1 Input from Lecture 7

Throughout, $\mathfrak{g}$ is a semisimple Lie algebra over $\mathbb{C}$ and κ is its Killing form. Fix a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$, that is, a maximal toral subalgebra. For $\lambda \in \mathfrak{h}^*$ put

$$\mathfrak{g}_\lambda = \{x \in \mathfrak{g} : [h, x] = \lambda(h)x \text{ for all } h \in \mathfrak{h}\}.$$

A *root* is a nonzero λ with $\mathfrak{g}_\lambda \neq 0$. Write R for the finite set of roots.

From Lecture 7 we use the root decomposition

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha.$$

We also use the bracket rule and orthogonality. For all $\lambda, \mu \in \mathfrak{h}^*$,

$$[\mathfrak{g}_\lambda, \mathfrak{g}_\mu] \subseteq \mathfrak{g}_{\lambda+\mu}, \qquad \kappa(\mathfrak{g}_\lambda, \mathfrak{g}_\mu) = 0 \quad \text{unless } \lambda + \mu = 0.$$

The Cartan subalgebra is abelian. The single cited import is the following.

Theorem 8.1 (Self-centraliser theorem · *C*)**.** *Let $\mathfrak{g}$ be a finite-dimensional semisimple Lie algebra over $\mathbb{C}$, and let $\mathfrak{h}$ be a maximal toral subalgebra. Then*

$$\mathfrak{g}_0 = C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}.$$

Source: Humphreys, Section 8.2, Proposition.

The other tools are Lie's theorem (Lecture 3), nondegeneracy of the Killing form (Lecture 4), complete reducibility (Lecture 5) and the $\mathfrak{sl}_2$-modules V_m (Lecture 6).

Lemma 8.2 (First consequences · *P*)**.** *(a) The restriction of κ to $\mathfrak{h}$ is nondegenerate.*
(b) For $\alpha \in R$, the pairing $\kappa : \mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha} \to \mathbb{C}$ is nondegenerate. In particular, $-\alpha \in R$.

(c) $Z(\mathfrak{g}) = 0$, *and* R *spans* $\mathfrak{h}^*$.

Proof. For $\mathfrak{g} = 0$ all three claims are immediate. Otherwise they are the orthogonality and root-spanning results proved in Lecture 7. That lecture also proved $Z(\mathfrak{g}) = 0$ from semisimplicity. ∎

By Lemma 8.2(a), each $\lambda \in \mathfrak{h}^*$ has a unique $t_\lambda \in \mathfrak{h}$ with

$$\kappa(t_\lambda, h) = \lambda(h) \quad \text{for all } h \in \mathfrak{h}.$$

The map $\lambda \mapsto t_\lambda$ is a linear isomorphism $\mathfrak{h}^* \to \mathfrak{h}$. Transport κ to $\mathfrak{h}^*$ by

$$(\lambda, \mu) = \kappa(t_\lambda, t_\mu) = \lambda(t_\mu) = \mu(t_\lambda).$$

This is a nondegenerate symmetric complex bilinear form.

8.2 The triple attached to a root

Lemma 8.3 (Opposite root vectors · *P*). *Let $\alpha \in R$, $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_{-\alpha}$. Then*

$$[x, y] = \kappa(x, y)\, t_\alpha.$$

Proof. The bracket rule and the self-centraliser theorem give $[x, y] \in \mathfrak{g}_0 = \mathfrak{h}$. For $h \in \mathfrak{h}$, invariance of κ gives

$$\kappa(h, [x, y]) = \kappa([h, x], y) = \alpha(h)\,\kappa(x, y) = \kappa\big(h, \kappa(x, y)\, t_\alpha\big).$$

Both $[x, y]$ and $\kappa(x, y)t_\alpha$ lie in $\mathfrak{h}$. By nondegeneracy on $\mathfrak{h}$ (Lemma 8.2(a)), they are equal. ∎

Lemma 8.4 (Roots are not isotropic · *P*). *For every $\alpha \in R$,*

$$(\alpha, \alpha) = \alpha(t_\alpha) \neq 0.$$

Proof. Suppose $\alpha(t_\alpha) = 0$. By Lemma 8.2(b), choose $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_{-\alpha}$ with $\kappa(x, y) = 1$. Put $s = [x, y]$. Lemma 8.3 gives $s = t_\alpha$, so

$$[s, x] = \alpha(s)x = 0, \qquad [s, y] = -\alpha(s)y = 0.$$

Thus $S = \operatorname{span}\{x, y, s\}$ is a subalgebra with $[S, S] = \mathbb{C}s$ and $[[S, S], [S, S]] = 0$. It is solvable.

The image $\operatorname{ad} S \subseteq \mathfrak{gl}(\mathfrak{g})$ is solvable as well. By Lie's theorem there is a basis of $\mathfrak{g}$ in which every $\operatorname{ad} z$, $z \in S$, is upper triangular. Then

$$\operatorname{ad} s = [\operatorname{ad} x, \operatorname{ad} y]$$

is strictly upper triangular, hence nilpotent.

On the other hand $s \in \mathfrak{h}$. By the root decomposition, $\operatorname{ad} s$ acts on $\mathfrak{g}_\gamma$ by $\gamma(s)$ and on $\mathfrak{h}$ by 0. So $\operatorname{ad} s$ is diagonalisable. A diagonalisable nilpotent operator is zero, so $s \in Z(\mathfrak{g}) = 0$. This contradicts $t_\alpha \neq 0$, which holds since $\alpha \neq 0$. ∎

Definition 8.5 (Coroot and root triple). Let $\alpha \in R$. The *coroot* of α is

$$h_\alpha = \alpha^\vee = \frac{2\, t_\alpha}{(\alpha, \alpha)} \in \mathfrak{h}.$$

For $\lambda \in \mathfrak{h}^*$ write $\langle \lambda, \alpha^\vee \rangle = \lambda(h_\alpha)$. A *root triple* for α is a triple $(x_\alpha, h_\alpha, y_\alpha)$ with $x_\alpha \in \mathfrak{g}_\alpha$, $y_\alpha \in \mathfrak{g}_{-\alpha}$ and $[x_\alpha, y_\alpha] = h_\alpha$.

Root triples exist. Choose $x_\alpha \neq 0$ in $\mathfrak{g}_\alpha$. Lemma 8.2(b) gives $y \in \mathfrak{g}_{-\alpha}$ with $\kappa(x_\alpha, y) \neq 0$. Rescale y to y_α with

$$\kappa(x_\alpha, y_\alpha) = \frac{2}{(\alpha, \alpha)}.$$

Lemma 8.3 then gives $[x_\alpha, y_\alpha] = h_\alpha$.

Proposition 8.6 (The rank-one subalgebra · *P*)**.** *Let $(x_\alpha, h_\alpha, y_\alpha)$ be a root triple. Then*

$$\alpha(h_\alpha) = 2, \qquad [h_\alpha, x_\alpha] = 2x_\alpha, \qquad [h_\alpha, y_\alpha] = -2y_\alpha.$$

Hence $S_\alpha = \operatorname{span}\{x_\alpha, h_\alpha, y_\alpha\}$ is a subalgebra, and $e \mapsto x_\alpha$, $h \mapsto h_\alpha$, $f \mapsto y_\alpha$ defines an isomorphism $\mathfrak{sl}_2 \to S_\alpha$. Moreover,

$$[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{C}h_\alpha,$$

and h_α is the only element of this line on which α takes the value 2.

Proof. First,

$$\alpha(h_\alpha) = \frac{2\,\alpha(t_\alpha)}{(\alpha, \alpha)} = 2.$$

Since $x_\alpha \in \mathfrak{g}_\alpha$, we get $[h_\alpha, x_\alpha] = \alpha(h_\alpha)x_\alpha = 2x_\alpha$. Similarly $[h_\alpha, y_\alpha] = -2y_\alpha$.

The relation $[x_\alpha, y_\alpha] = h_\alpha \neq 0$ forces $x_\alpha \neq 0$ and $y_\alpha \neq 0$. The three vectors lie in the independent summands $\mathfrak{g}_\alpha$, $\mathfrak{h}$ and $\mathfrak{g}_{-\alpha}$, so they are linearly independent. The linear map $\mathfrak{sl}_2 \to S_\alpha$ is therefore bijective. It respects the three defining brackets of $\mathfrak{sl}_2$, which determine all brackets. So it is an isomorphism onto S_α.

Finally, Lemma 8.3 gives $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \subseteq \mathbb{C}t_\alpha = \mathbb{C}h_\alpha$, and the line contains h_α. For $c \in \mathbb{C}$, $\alpha(ch_\alpha) = 2c$, which equals 2 only for $c = 1$. ∎

Lecture 6 for S_α

Through the isomorphism, every finite-dimensional $\mathfrak{g}$-module V, such as $\mathfrak{g}$ under ad, is an S_α-module. Write

$$V(k) = \{v \in V : h_\alpha v = kv\}.$$

By Lecture 5 and Lecture 6, V is a direct sum of modules V_m. In V_m the basis vectors $v_j = f^j v_0$, $0 \leq j \leq m$, satisfy

$$hv_j = (m - 2j)v_j, \qquad ev_j = j(m - j + 1)v_{j-1}.$$

We use three consequences.

(W1) h_α is diagonalisable on V, with integer eigenvalues.

(W2) The number of summands V_m equals $\dim V(0) + \dim V(1)$. Indeed, each V_m contains exactly one of the eigenvalues $0, 1$, with multiplicity one.

(W3) In V_m the eigenvalues are $m, m-2, \ldots, -m$, each of multiplicity one. For $-m \leq k < m$, the operator x_α maps $V_m(k)$ isomorphically onto $V_m(k+2)$, since $j(m - j + 1) \neq 0$ for $1 \leq j \leq m$.

8.3 Root spaces are lines

Theorem 8.7 (Dimension one and reducedness · *P*)**.** *Let $\alpha \in R$.*

(a) $\dim \mathfrak{g}_\alpha = 1$.

(b) If $c \in \mathbb{C}$ and $c\alpha \in R$, then $c = \pm 1$.

Proof. Fix a root triple for α. On $\mathfrak{g}_{c\alpha}$ the coroot acts by $c\alpha(h_\alpha) = 2c$. Consider

$$M = \mathfrak{h} \oplus \bigoplus_{c \neq 0,\ c\alpha \in R} \mathfrak{g}_{c\alpha}.$$

Step 1: M is an S_α-submodule. The operator $\operatorname{ad} h_\alpha$ preserves each summand. By the bracket rule, $\operatorname{ad} x_\alpha$ sends $\mathfrak{g}_{c\alpha}$ into $\mathfrak{g}_{(c+1)\alpha}$. This is $\mathfrak{h}$ when $c = -1$, and otherwise it is zero or a summand of M. Also $\operatorname{ad} x_\alpha$ sends $\mathfrak{h}$ into $\mathfrak{g}_\alpha$. The argument for $\operatorname{ad} y_\alpha$ is the same.

Step 2: two known submodules. The subalgebra $S_\alpha \subseteq M$ is a submodule isomorphic to V_2. The hyperplane $\ker \alpha \subseteq \mathfrak{h}$ is a trivial submodule: for $k \in \ker \alpha$,

$$[x_\alpha, k] = -\alpha(k)x_\alpha = 0, \qquad [y_\alpha, k] = \alpha(k)y_\alpha = 0, \qquad [h_\alpha, k] = 0.$$

Since $\alpha(h_\alpha) = 2$, we have $\mathfrak{h} = \ker \alpha \oplus \mathbb{C}h_\alpha$ and $\ker \alpha \cap S_\alpha = 0$. Complete reducibility gives a submodule M' with

$$M = \ker \alpha \oplus S_\alpha \oplus M'.$$

Step 3: M' has only odd eigenvalues. The eigenvalue 0 occurs on M only for $c = 0$. Hence

$$M(0) = \mathfrak{h} = \ker \alpha \oplus \mathbb{C}h_\alpha,$$

and $M'(0) = 0$. Every V_m with m even has the eigenvalue 0. So every summand of M' is some V_m with m odd.

Step 4: $2\alpha \notin R$. The eigenspace $M(4)$ is $\mathfrak{g}_{2\alpha}$. The summands $\ker \alpha$ and S_α have eigenvalues $0, \pm 2$ only, and M' has odd eigenvalues. Thus $M(4) = 0$, and $2\alpha \notin R$. This holds for every root.

Step 5: $M' = 0$. If $\alpha/2$ were a root, Step 4 applied to $\alpha/2$ would give $\alpha \notin R$. So $\alpha/2 \notin R$. The eigenvalue 1 on M could only come from $c = 1/2$, hence

$$M(1) = 0.$$

Every V_m with m odd has the eigenvalue 1. Hence $M' = 0$.

Conclusion. Now $M = \ker \alpha \oplus S_\alpha$, with eigenvalues $0, \pm 2$ only. For $c \neq 0$ this forces $2c = \pm 2$, which is (b). The eigenspace $M(2)$ equals $\mathfrak{g}_\alpha$ and also $\mathbb{C}x_\alpha$. This is (a). ∎

8.4 Root strings and integrality

Definition 8.8 (Root string)**.** Let $\alpha, \beta \in R$ with $\beta \neq \pm\alpha$. The *α-string through β* is the set of roots of the form $\beta + i\alpha$ with $i \in \mathbb{Z}$.

Theorem 8.9 (Root strings · *P*). *Let $\alpha, \beta \in R$ with $\beta \neq \pm\alpha$.*

(a) $\langle \beta, \alpha^\vee \rangle \in \mathbb{Z}$.

(b) There are integers $r, q \geq 0$ such that, for $i \in \mathbb{Z}$,

$$\beta + i\alpha \in R \iff -r \leq i \leq q.$$

(c) These integers satisfy

$$r - q = \langle \beta, \alpha^\vee \rangle.$$

(d) If $\alpha + \beta \in R$, then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$.

Since $\langle \pm\alpha, \alpha^\vee \rangle = \pm 2$, part (a) makes every $\langle \beta, \alpha^\vee \rangle$ with $\alpha, \beta \in R$ an integer, a *Cartan integer*.

Proof. Fix a root triple for α and put

$$K = \bigoplus_{i \in \mathbb{Z}} \mathfrak{g}_{\beta+i\alpha}.$$

No $\beta + i\alpha$ is zero. Otherwise $\beta = -i\alpha$ would be a root proportional to α, and Theorem 8.7(b) would give $\beta = \pm\alpha$. By the bracket rule, $\operatorname{ad} x_\alpha$ and $\operatorname{ad} y_\alpha$ move each summand to a neighbouring summand, never into $\mathfrak{h}$. Thus K is an S_α-submodule. It is nonzero because $\mathfrak{g}_\beta \neq 0$.

The coroot acts on $\mathfrak{g}_{\beta+i\alpha}$ by

$$(\beta + i\alpha)(h_\alpha) = \langle \beta, \alpha^\vee \rangle + 2i.$$

By (W1) this is an integer. The case $i = 0$ gives (a).

By Theorem 8.7(a), each summand has dimension at most one. The eigenvalues $\langle \beta, \alpha^\vee \rangle + 2i$ are distinct and have the same parity. Hence $\dim K(0) + \dim K(1) \leq 1$, and (W2) shows that K is irreducible, say $K \cong V_m$. By (W3), the eigenvalues on K are exactly $m, m-2, \ldots, -m$. The value for $i = 0$ is one of them, so every value $\langle \beta, \alpha^\vee \rangle + 2i$ has the parity of m. Hence $\beta + i\alpha \in R$ exactly when

$$-m \leq \langle \beta, \alpha^\vee \rangle + 2i \leq m.$$

These i form an interval of integers containing 0. Call it $[-r, q]$. This is (b).

The endpoints give the highest and lowest eigenvalues:

$$\langle \beta, \alpha^\vee \rangle + 2q = m, \qquad \langle \beta, \alpha^\vee \rangle - 2r = -m.$$

Adding the two equations gives $2\langle \beta, \alpha^\vee \rangle = 2r - 2q$, which is (c).

For (d), $\alpha + \beta \in R$ means $q \geq 1$. Then $k = \langle \beta, \alpha^\vee \rangle = m - 2q < m$, and $k \geq -m$. By (W3), $\operatorname{ad} x_\alpha$ maps $\mathfrak{g}_\beta = K(k)$ isomorphically onto $K(k+2) = \mathfrak{g}_{\alpha+\beta}$. Since $\mathfrak{g}_\alpha = \mathbb{C}x_\alpha$ by Theorem 8.7(a),

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = [x_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}.$$

■

For $\alpha \in R$, define the linear map

$$s_\alpha : \mathfrak{h}^* \longrightarrow \mathfrak{h}^*, \qquad s_\alpha(\lambda) = \lambda - \lambda(h_\alpha)\alpha.$$

Since $\alpha(h_\alpha) = 2$, we have $s_\alpha(\lambda)(h_\alpha) = -\lambda(h_\alpha)$. Applying the formula again gives $s_\alpha^2 = \mathrm{id}$.

Corollary 8.10 (Reflections permute the roots · *P*). *For $\alpha, \beta \in R$,*

$$s_\alpha(\beta) = \beta - \langle \beta, \alpha^\vee \rangle \alpha \in R.$$

Proof. For $\beta = \pm\alpha$ the element is $\mp\alpha$, a root by Lemma 8.2(b). Otherwise it equals $\beta + (q - r)\alpha$, which is a root because $-r \leq q - r \leq q$. Thus $s_\alpha(R) \subseteq R$. Since $s_\alpha^2 = \mathrm{id}$, equality holds. ■

8.5 The real span of the roots

Lemma 8.11 (The Killing form on $\mathfrak{h} \cdot P$). *For $h, h' \in \mathfrak{h}$ and $\lambda, \mu \in \mathfrak{h}^*$,*

$$\kappa(h, h') = \sum_{\gamma \in R} \gamma(h)\gamma(h'), \qquad (\lambda, \mu) = \sum_{\gamma \in R} (\gamma, \lambda)(\gamma, \mu).$$

Proof. The operator $\operatorname{ad} h \operatorname{ad} h'$ is 0 on $\mathfrak{h}$ and multiplication by $\gamma(h)\gamma(h')$ on the line $\mathfrak{g}_\gamma$. Take its trace. For the second formula, put $h = t_\lambda$, $h' = t_\mu$ and use $\gamma(t_\lambda) = (\gamma, \lambda)$. ∎

Theorem 8.12 (The Euclidean space of roots · *P*). *(a) For $\alpha, \beta \in R$, $(\alpha, \beta) \in \mathbb{Q}$ and $(\alpha, \alpha) > 0$.*

(b) Let E be the real span of R in $\mathfrak{h}^*$. Then $\dim_{\mathbb{R}} E = \dim_{\mathbb{C}} \mathfrak{h}^*$ and $\mathfrak{h}^* = E \oplus iE$.

(c) The form $(\ ,\ )$ is real-valued and positive definite on E.

Proof. (a) Lemma 8.11 gives $(\beta, \beta) = \sum_\gamma (\gamma, \beta)^2$. Divide by $(\beta, \beta)^2$, which is nonzero by Lemma 8.4. Since $\langle \gamma, \beta^\vee \rangle = 2(\gamma, \beta)/(\beta, \beta)$,

$$\frac{1}{(\beta, \beta)} = \sum_{\gamma \in R} \left(\frac{(\gamma, \beta)}{(\beta, \beta)} \right)^2 = \frac{1}{4} \sum_{\gamma \in R} \langle \gamma, \beta^\vee \rangle^2.$$

The right side is a quarter of a sum of squares of integers, and its term $\gamma = \beta$ equals 1. So (β, β) is a positive rational. Then $(\gamma, \beta) = \frac{1}{2}\langle \gamma, \beta^\vee \rangle (\beta, \beta)$ is rational.

(b) By Lemma 8.2(c), choose roots $\alpha_1, \dots, \alpha_\ell$ forming a basis of $\mathfrak{h}^*$. Their Gram matrix $G = ((\alpha_i, \alpha_j))$ is rational by (a) and invertible, because the form is nondegenerate. For $\beta \in R$ write $\beta = \sum_i c_i \alpha_i$ with $c_i \in \mathbb{C}$. Then

$$(\beta, \alpha_j) = \sum_{i=1}^{\ell} c_i (\alpha_i, \alpha_j) \quad (1 \le j \le \ell).$$

The left sides are rational, so inverting G shows that every c_i is rational. Thus $E = \mathbb{R}\alpha_1 \oplus \cdots \oplus \mathbb{R}\alpha_\ell$. Since the α_i form a $\mathbb{C}$-basis, $\mathfrak{h}^* = E \oplus iE$.

(c) The form is bilinear and, by (a), real on pairs of roots. Since E is the real span of R, the value (λ, μ) is real for all $\lambda, \mu \in E$. For $\lambda \in E$, Lemma 8.11 gives

$$(\lambda, \lambda) = \sum_{\gamma \in R} (\gamma, \lambda)^2 \ge 0.$$

If equality holds, then $(\gamma, \lambda) = 0$ for every root. Since R spans $\mathfrak{h}^*$ and the form is nondegenerate, $\lambda = 0$. ∎

For $\alpha \in R$ and $\lambda \in E$, the reflection formula becomes

$$s_\alpha(\lambda) = \lambda - \frac{2(\lambda, \alpha)}{(\alpha, \alpha)} \alpha.$$

The coefficient is real, so s_α maps E to itself.

Positive definiteness gives an orthogonal decomposition $E = \mathbb{R}\alpha \oplus \alpha^\perp$. The map s_α fixes $\alpha^\perp$ and negates α, so it is an orthogonal reflection of E. By Corollary 8.10, it permutes R.

The *Weyl group* $W \subseteq \mathrm{GL}(E)$ is generated by all s_α. The form on E is W-invariant. Lecture 9 studies W.

Remark 8.13 (The complex Killing form is not Euclidean)**.** The form is positive definite on the real span E. On $\mathfrak{h}^*$ it is complex bilinear, so $(i\lambda, i\lambda) = -(\lambda, \lambda)$. On $\mathfrak{g}$ itself, since $2\alpha \neq 0$, orthogonality gives $\kappa(x_\alpha, x_\alpha) = 0$ although $x_\alpha \neq 0$.

In this lecture the form $(\ ,\ )$ is always the one induced by the Killing form. The size of its values depends on that choice of scale. Suppose that we replace κ by $c\kappa$ with $c > 0$. Then t_λ becomes $c^{-1}t_\lambda$, and (λ, μ) becomes $c^{-1}(\lambda, \mu)$. The coroot $h_\alpha = 2t_\alpha/(\alpha, \alpha)$ does not change. Hence the Cartan integers, the reflections, the angles and the ratios of lengths do not change either. Lecture 9 therefore allows any positive multiple of the form. Lecture 16 chooses the multiple on each simple ideal so that long roots have squared length 2.

8.6 Root strings in A_2 and C_2

Write E_{ij} for matrix units, and ε_i for the i-th diagonal entry as a linear form on diagonal matrices. For roots α_1, α_2 with coroots h_1, h_2 put

$$a_{ij} = \alpha_j(h_i).$$

The matrix (a_{ij}) is called the *Cartan matrix* of the pair, and Lecture 10 uses the same convention. Lecture 9 shows that the pairs below are simple roots.

Figure 8.1 previews the strings that the two examples below compute. The examples do not use the figure. Each dot is a root. Each number is the value at that root of the coroot that acts on the string. A string with $m+1$ points is a copy of V_m, so these values run through $-m, -m+2, \ldots, m$. In A_2 the two strings are exchanged by renumbering, and each has two points. In C_2 the α_1-string through α_2 has three points, and the α_2-string through α_1 has two points. This difference produces the two different entries a_{12} and a_{21}.

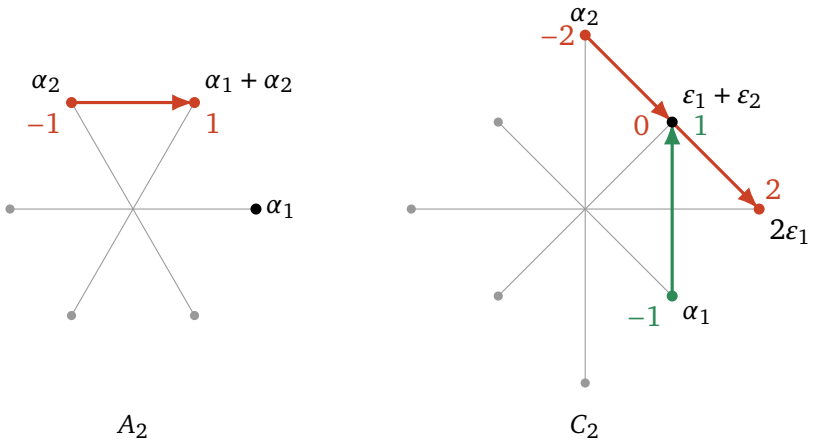


Figure 8.1: Root strings in A_2 and C_2. Red arrows trace α_1-strings through α_2, with the values of h_1. The green arrow is the α_2-string through α_1, with the values of h_2.

The scale of each panel is chosen for display. Only angles and ratios of lengths in a panel carry meaning. In the C_2 panel the point (x, y) is $x\varepsilon_1 + y\varepsilon_2$, and the plotted lengths use the dot product of coefficients. Exercise 8.3(i) shows that the form $(\ ,\ )$ on these roots is a positive multiple of this dot product, and it computes the multiple. In this plot long roots have squared length 4. That number is neither the Killing value nor the normalisation of Lecture 16. In the A_2 panel every root has squared length 2, the scale that Lecture 9 uses for A_2.

Example 8.14 ($A_2 = \mathfrak{sl}_3$)**.** Use the traceless diagonal Cartan subalgebra of $\mathfrak{sl}_3$ from Lecture 7. Its roots are $\varepsilon_i - \varepsilon_j$ for $i \neq j$. For $\alpha = \varepsilon_i - \varepsilon_j$,

$$[E_{ij}, E_{ji}] = E_{ii} - E_{jj}, \qquad \alpha(E_{ii} - E_{jj}) = 2.$$

By Proposition 8.6, $h_\alpha = E_{ii} - E_{jj}$.

Put $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = \varepsilon_2 - \varepsilon_3$. Test the α_1-string through α_2:

$$\begin{aligned} \alpha_2 - \alpha_1 &= -\varepsilon_1 + 2\varepsilon_2 - \varepsilon_3 \notin R, \\ \alpha_2 + \alpha_1 &= \varepsilon_1 - \varepsilon_3 \in R, \\ \alpha_2 + 2\alpha_1 &= 2\varepsilon_1 - \varepsilon_2 - \varepsilon_3 \notin R. \end{aligned}$$

The coefficient vectors in this test all have sum zero. Suppose that two such vectors c and d define the same functional on $\mathfrak{h}$. Then $c - d = t(1,1,1)$ for some $t \in \mathbb{C}$, because $\mathfrak{h}$ consists of the traceless diagonal matrices. Summing the coordinates gives $3t = 0$, so $c = d$. The two excluded forms are therefore not roots. Thus $r = 0$ and $q = 1$. The string theorem gives

$$a_{12} = \alpha_2(h_1) = r - q = -1.$$

Directly, $\alpha_2(E_{11} - E_{22}) = -1 - 0 = -1$. Exchanging α_1 and α_2 gives $a_{21} = -1$.

Example 8.15 ($C_2 = \mathfrak{sp}_4$)**.** Let

$$J = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}, \qquad \mathfrak{sp}_4 = \{X \in \mathfrak{gl}_4(\mathbb{C}) : X^{\mathsf{T}} J + JX = 0\}.$$

Solving the condition in 2×2 blocks gives

$$X = \begin{pmatrix} A & B \\ C & -A^{\mathsf{T}} \end{pmatrix}, \qquad B = B^{\mathsf{T}}, \quad C = C^{\mathsf{T}}.$$

Exercise 8.1 proves that $\mathfrak{sp}_4$ is semisimple and that the matrices $\operatorname{diag}(a_1, a_2, -a_1, -a_2)$ form a Cartan subalgebra $\mathfrak{h}$. Computing $[h, X]$ gives the roots and root vectors below. In each line, the first vector belongs to the plus sign:

$$\begin{aligned} \pm(\varepsilon_1 - \varepsilon_2): &\quad E_{12} - E_{43}, \quad E_{21} - E_{34}, \\ \pm(\varepsilon_1 + \varepsilon_2): &\quad E_{14} + E_{23}, \quad E_{41} + E_{32}, \\ \pm 2\varepsilon_1: &\quad E_{13}, \quad E_{31}, \\ \pm 2\varepsilon_2: &\quad E_{24}, \quad E_{42}. \end{aligned}$$

Put $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = 2\varepsilon_2$. Brackets of opposite root vectors give

$$\begin{aligned} [E_{12} - E_{43}, E_{21} - E_{34}] &= E_{11} - E_{22} - E_{33} + E_{44}, \\ [E_{24}, E_{42}] &= E_{22} - E_{44}. \end{aligned}$$

Each root takes the value 2 on the corresponding right side. By Proposition 8.6,

$$h_1 = \operatorname{diag}(1, -1, -1, 1), \qquad h_2 = \operatorname{diag}(0, 1, 0, -1).$$

The α_1-string through α_2 is found by testing:

$$\begin{aligned} \alpha_2 - \alpha_1 &= -\varepsilon_1 + 3\varepsilon_2 \notin R, \\ \alpha_2 + \alpha_1 &= \varepsilon_1 + \varepsilon_2 \in R, \\ \alpha_2 + 2\alpha_1 &= 2\varepsilon_1 \in R, \\ \alpha_2 + 3\alpha_1 &= 3\varepsilon_1 - \varepsilon_2 \notin R. \end{aligned}$$

Thus $r = 0$ and $q = 2$, and

$$a_{12} = \alpha_2(h_1) = r - q = -2.$$

This agrees with the direct value $2\varepsilon_2(h_1) = 2 \cdot (-1)$.

The α_2-string through α_1 is shorter:

$$\alpha_1 - \alpha_2 = \varepsilon_1 - 3\varepsilon_2 \notin R,$$
$$\alpha_1 + \alpha_2 = \varepsilon_1 + \varepsilon_2 \in R,$$
$$\alpha_1 + 2\alpha_2 = \varepsilon_1 + 3\varepsilon_2 \notin R.$$

So $r = 0$, $q = 1$ and $a_{21} = \alpha_1(h_2) = -1$. The Cartan matrix of this pair is

$$(a_{ij}) = \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}.$$

The asymmetry of this matrix records root lengths. By definition, $a_{ij} = \alpha_j(h_i) = 2(\alpha_j, \alpha_i)/(\alpha_i, \alpha_i)$. Since $(\alpha_1, \alpha_2) \neq 0$,

$$\frac{a_{12}}{a_{21}} = \frac{(\alpha_2, \alpha_2)}{(\alpha_1, \alpha_1)} = 2.$$

So α_2 is long and α_1 is short. Reflections transport lengths:

$$s_{\alpha_2}(\alpha_1) = \alpha_1 - a_{21}\alpha_2 = \varepsilon_1 + \varepsilon_2, \qquad s_{\alpha_1}(\alpha_2) = \alpha_2 - a_{12}\alpha_1 = 2\varepsilon_1.$$

By W-invariance, the roots $\pm\varepsilon_1 \pm \varepsilon_2$ are short and the roots $\pm 2\varepsilon_i$ are long.

Self-checks

(1) In $\mathfrak{sl}_3$, find the $(\varepsilon_1 - \varepsilon_2)$-string through $\varepsilon_1 - \varepsilon_3$, and the value of $\varepsilon_1 - \varepsilon_3$ on $h_{\varepsilon_1 - \varepsilon_2}$.

(2) Why is the Cartan matrix of C_2 not symmetric, while that of A_2 is?

(3) For $\alpha \in R$, compute $\kappa(ih_\alpha, ih_\alpha)$.

Answers. (1) Put $\alpha = \varepsilon_1 - \varepsilon_2$ and $\beta = \varepsilon_1 - \varepsilon_3$. Then $\beta - \alpha = \varepsilon_2 - \varepsilon_3 \in R$, while $\beta - 2\alpha$ and $\beta + \alpha$ are not roots, by the same coefficient test. So $r = 1$, $q = 0$ and the value is 1. (2) The ratio a_{12}/a_{21} equals a ratio of squared lengths. It is 2 in C_2 and 1 in A_2. (3) Since $\kappa(t_\alpha, t_\alpha) = (\alpha, \alpha) > 0$,

$$\kappa(ih_\alpha, ih_\alpha) = -\frac{4\,\kappa(t_\alpha, t_\alpha)}{(\alpha, \alpha)^2} = -\frac{4}{(\alpha, \alpha)} < 0.$$

Summary

Each root α gives $S_\alpha \cong \mathfrak{sl}_2$. Viewing $\mathfrak{g}$ as an S_α-module shows that root spaces are lines and that $\pm\alpha$ are the only roots proportional to α.

For $\beta \neq \pm\alpha$, the root spaces of the α-string through β span an irreducible S_α-module, and $r - q = \langle \beta, \alpha^\vee \rangle$. The real span E of the roots carries a positive definite W-invariant form. Lecture 9 axiomatises this structure.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Sections 8.3–8.5. Erdmann–Wildon, *Introduction to Lie Algebras*, Sections 10.4–10.6.